

Lab 2 Writeup By Dalton JS Crawford

The Ping Pong algorithm sends and receives information from one leading node. Each additional node waits for a message from the starting node, does its work, and then sends it back. The send and receive commands MPI_Send and MPI_Recv are blocking so, each additional node must wait on the original node to send its request before work can begin. This feature can be optional in the architecture of this algorithm depending on the kind of work you are looking to do, but for this exercise we left it blocking.

The Ring algorithm sends information in a ring, from one node to the next. The origin node sends to the first node, and then waits for the final node to send information to it. Each subsequent node receives from the previous node and sends to the next. MPI_Send and MPI_Recv are blocking. This is required for the ring architecture to work at all.

In calculating the bandwidth and analyzing the performance of these algorithms, the average time for the entire algorithm to run was taken, including all the send and receives, one assignment, and no other work. The Average time per hop was calculated differently for each algorithm because they have a different number of hops. The ping pong algorithm has $(n-1)*2$ hops where n is the rank or number of processors used. The ring is n hops where n is the rank or number of processors used. The number of bytes transferred in the experiments is 1024^3 bytes. In the graph this number is simplified to 1 Gb. The bandwidth is then $1\text{Gb}/\text{average time per hop}$.

